

RAG Retrieval Pipeline Benchmark Report

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System / Setting	Dataset	Threshold	Hit-Rate@1	Hit-Rate@3	MRR	Abstention	Fabrication	Accuracy	Latency
BM25	Eval	0.00	92.00%	100.00%	0.960	0.00%	100.00%	65.71%	0.1 ms
BM25	Test	0.00	88.89%	94.44%	0.927	0.00%	100.00%	80.00%	0.1 ms
RETRIEVER	Eval	0.50	96.00%	100.00%	0.980	10.00%	90.00%	71.43%	13.2 ms
RETRIEVER	Eval	0.65	84.00%	88.00%	0.860	90.00%	10.00%	85.71%	13.7 ms
RETRIEVER	Test	0.50	88.89%	95.56%	0.922	10.00%	90.00%	81.00%	14.6 ms
RETRIEVER	Test	0.65	65.56%	70.00%	0.678	60.00%	40.00%	65.00%	14.7 ms
RETRIEVER (Sliding Window)	Eval	0.50	96.00%	100.00%	0.980	20.00%	80.00%	74.29%	13.6 ms
RETRIEVER (Sliding Window)	Eval	0.65	80.00%	80.00%	0.800	80.00%	20.00%	80.00%	13.7 ms
RETRIEVER (Sliding Window)	Test	0.50	90.00%	93.33%	0.915	10.00%	90.00%	82.00%	14.8 ms
RETRIEVER (Sliding Window)	Test	0.65	63.33%	64.44%	0.639	60.00%	40.00%	63.00%	14.9 ms
RERANKER	Eval	0.50	96.00%	100.00%	0.980	10.00%	90.00%	71.43%	13.4 ms
RERANKER	Eval	0.65	84.00%	88.00%	0.860	90.00%	10.00%	85.71%	14.0 ms
RERANKER	Test	0.50	88.89%	95.56%	0.922	10.00%	90.00%	81.00%	14.7 ms
RERANKER	Test	0.65	65.56%	70.00%	0.678	60.00%	40.00%	65.00%	14.9 ms
RERANKER (Sliding Window)	Eval	0.50	100.00%	100.00%	1.000	20.00%	80.00%	77.14%	16.5 ms
RERANKER (Sliding Window)	Eval	0.65	80.00%	80.00%	0.800	80.00%	20.00%	80.00%	16.3 ms
RERANKER (Sliding Window)	Test	0.50	90.00%	92.22%	0.911	10.00%	90.00%	82.00%	17.4 ms
RERANKER (Sliding Window)	Test	0.65	63.33%	64.44%	0.639	60.00%	40.00%	63.00%	16.7 ms

Metric Descriptions & Evaluation Notes:

- **Hit-Rate@1 / Hit-Rate@3:** Rate of answerable queries where the correct document was retrieved at rank 1, or within the top 3 (deduplicated by document ID) respectively.
- **MRR (Mean Reciprocal Rank):** Reciprocal of the rank of the first correct document ($1/\text{rank}$), averaged over all answerable queries.
- **Abstention Rate:** The fraction of unanswerable queries where the system correctly returned no document (top-1 score was below threshold).
- **Fabrication Rate:** The fraction of unanswerable queries where the system incorrectly fabricated an answer (top-1 score met/exceeded threshold).
- **Accuracy:** The overall percentage of correct behaviors (correct retrievals for answerable queries + correct abstentions for unanswerable ones).
- **Latency:** The average retrieval / rerank execution time per query in milliseconds.